

ME-203A: Kinematics of Machines

by

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SYLLABUS:

UNIT 1

Kinematic analysis of mechanism: Definitions of Links, pairs and kinematic chain, Mechanism and machine, Inversion of four bar kinematic chain and slider crank chains.

Velocity and acceleration analysis: Relative velocity method; Instantaneous center method, Coriolis component of acceleration

UNIT 2

Special purpose mechanism: Mechanisms for straight line, Motorcar steering mechanisms, Hooks joint and Geneva mechanism

UNIT 3

Kinematics synthesis of mechanism: Type of synthesis, DOF, Grashof's criterion of movability, Limit position, Dead center transmission angle synthesis of mechanism by relative pole method, Inversion method, Freudenstein's methods of synthesis, Design of a four bar chain for constant angular velocity ratio of cranks.

UNIT 4

Friction: Solid, rolling greasy and viscous friction, Laws of Friction inclined plane, Friction of pivots and collars, Friction circle of turning pair and friction axis of a link, Ball and roller bearings, Single and multiple disc-clutches

UNIT 5

Belts, Ropes & Chains: Mechanics of belt drive, Length of belt, Ratio of tensions in flat and V belt, Effect of centrifugal force, Power transmission capacity, Belt materials.

Rope drive: Types of ropes, Construction, Forces on rope and breaking strength,

Chain drive: Types, Chain length, Chordal action hoisting and power transmission

ME 203 A: KINEMATICS OF MACHINES

2L, 1T

3 Hours, 100 Marks

COURSE OUTCOMES:

CO1: Describe the concepts of mechanisms, machines and related terminologies and discuss different types of methods to analyse the velocity and acceleration of various mechanism.

CO2: Discuss Geneva mechanism.

CO3: Design of a four bar chain for constant angular velocity ratio of cranks.

CO4: Explain the concepts of friction and its types. In addition, assess its effects of different mechanical components.

CO5: Analyze various motion transmission elements like belt drive, rope drive and chain drive.

UNIT 1

Kinematic analysis of mechanism: Definitions of Links, pairs and kinematic chain, Mechanism and machine, Inversion of four bar kinematic chain and slider crank chains.

Velocity and acceleration analysis: Relative velocity method; Instantaneous center method, Coriolis component of acceleration

Reference Books:

1. Theory of Machines, Kinematics and Dynamics, By Sadhu Singh, Published by Pearson.
2. Theory of Machines, By Thomas Bevan, Published by Pearson.
3. Theory of Machines, By S. S. Rattan, Published by Mc Graw Hill Education.
4. Theory of Machine, R.S. Khurmi and J.K. Gupta, Published by Eurasia Publishing House (Pvt.) Ltd.

Kinematic Chain:

A kinematic chain may be defined as an assembly of links in which the relative motion of the links is possible and the motion of each relative to the others is definite. The last link of the kinematic chain is attached to the first link. The four-bar mechanism and the slider crank mechanism are some of the examples of a kinematic chain.

➤ **Relationship for kinematic chain having lower pairs:**

$$L = 2P - 4$$

$$J = 3L/2 - 2$$

Where: L = number of binary links

P = number of lower pairs

J = number of binary joints

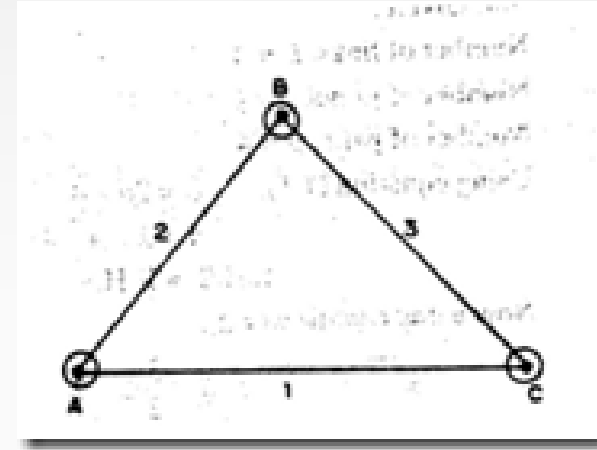
If $LHS > RHS$, then chain is called locked chain or redundant chain
 $LHS = RHS$, then chain is constrained
 $LHS < RHS$, then chain is unconstrained

- For a kinematic chain having higher pairs, each higher pair is taken equivalent to two lower pairs:

$$J + H/2 = 3L/2 - 2$$

Where H = number of higher pairs.

Example 1: A chain with three links is shown here.
Prove that the chain is locked.



Reference: researchgate.net

Solution: Number of binary joints, $J = 3$
Number of binary links, $L = 3$
Number of lower pairs, $P = 3$

Firstly, No. of lower links:

$$L = 2P - 4$$

$$3 = 2 \times 3 - 4 = 2$$

$$LHS > RHS$$

Secondly, No. of binary joints:

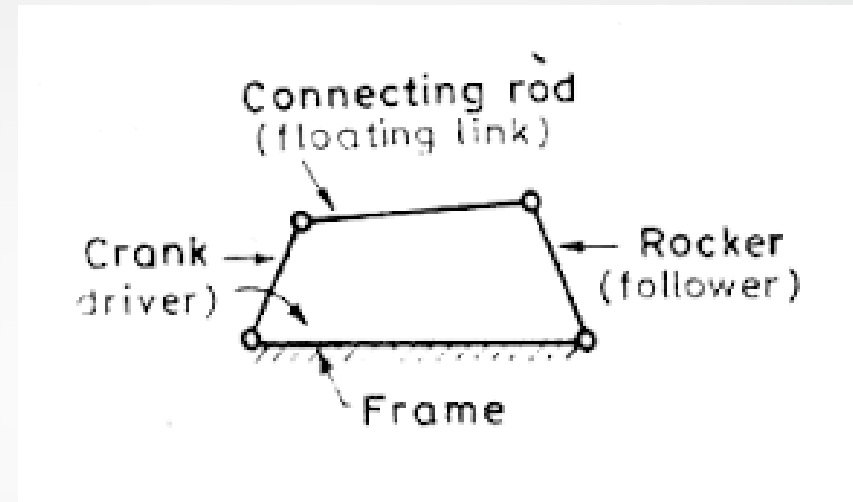
$$J = 3L/2 - 2$$

$$3 = 3 \times 3/2 - 2 = 2$$

$$LHS > RHS$$

Therefore, it is a locked chain.

Example 2: A four-bar chain is shown here. Prove that it is a constrained chain.



Reference: researchgate.net

Solution: Number of binary joints, $J = 4$
 Number of binary links, $L = 4$
 Number of lower pairs, $P = 4$

Firstly, No. of lower links:

$$L = 2P - 4$$

$$4 = 2 \times 4 - 4 = 4$$

$$\text{LHS} = \text{RHS}$$

Secondly, No. of binary joints:

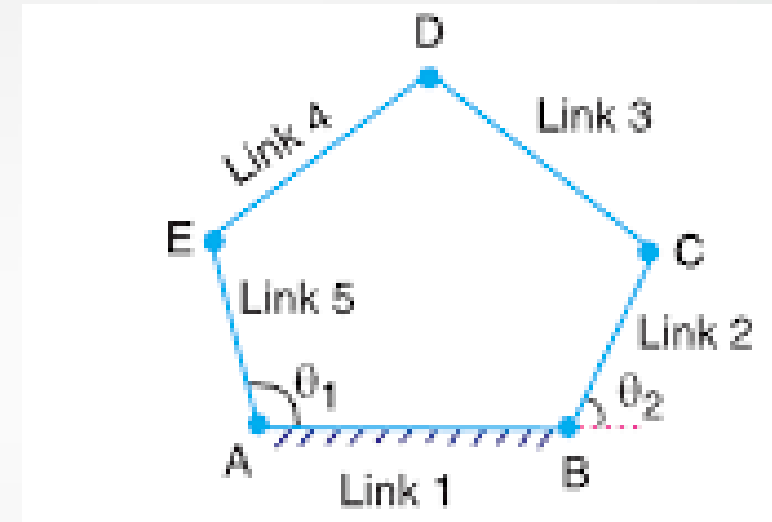
$$J = \frac{3L}{2} - 2$$

$$4 = \frac{3 \times 4}{2} - 2 = 4$$

$$\text{LHS} = \text{RHS}$$

Therefore, it is a constrained chain.

Example 3: A five-bar chain is shown here. Prove that it is an unconstrained chain.



Reference: researchgate.net

Solution: Number of binary joints, $J = 5$
 Number of binary links, $L = 5$
 Number of lower pairs, $P = 5$

Firstly, No. of lower links:

$$L = 2P - 4$$

$$5 = 2 \times 5 - 4 = 6$$

$$\text{LHS} < \text{RHS}$$

Secondly, No. of binary joints:

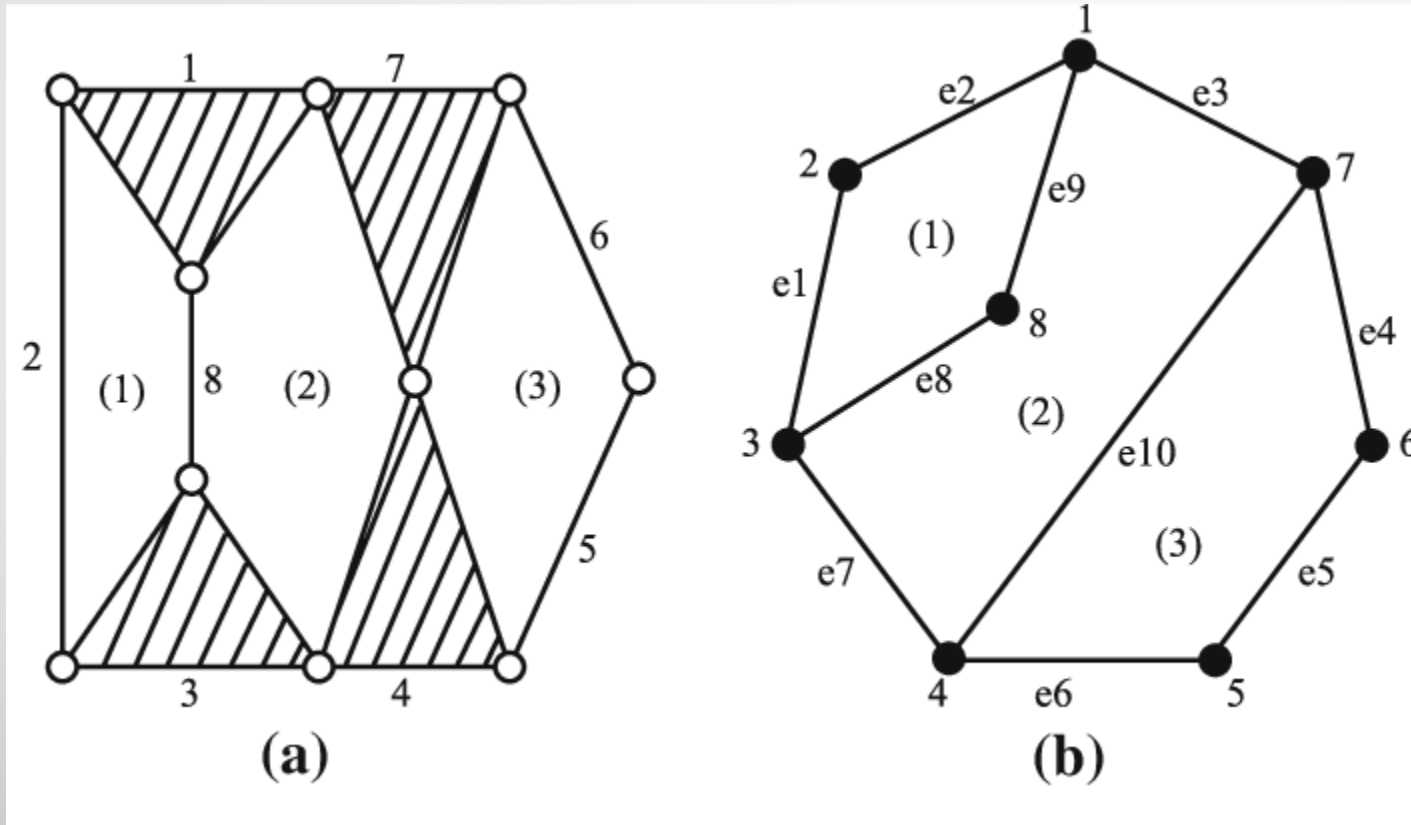
$$J = \frac{3L}{2} - 2$$

$$5 = \frac{3 \times 5}{2} - 2 = 5.5$$

$$\text{LHS} < \text{RHS}$$

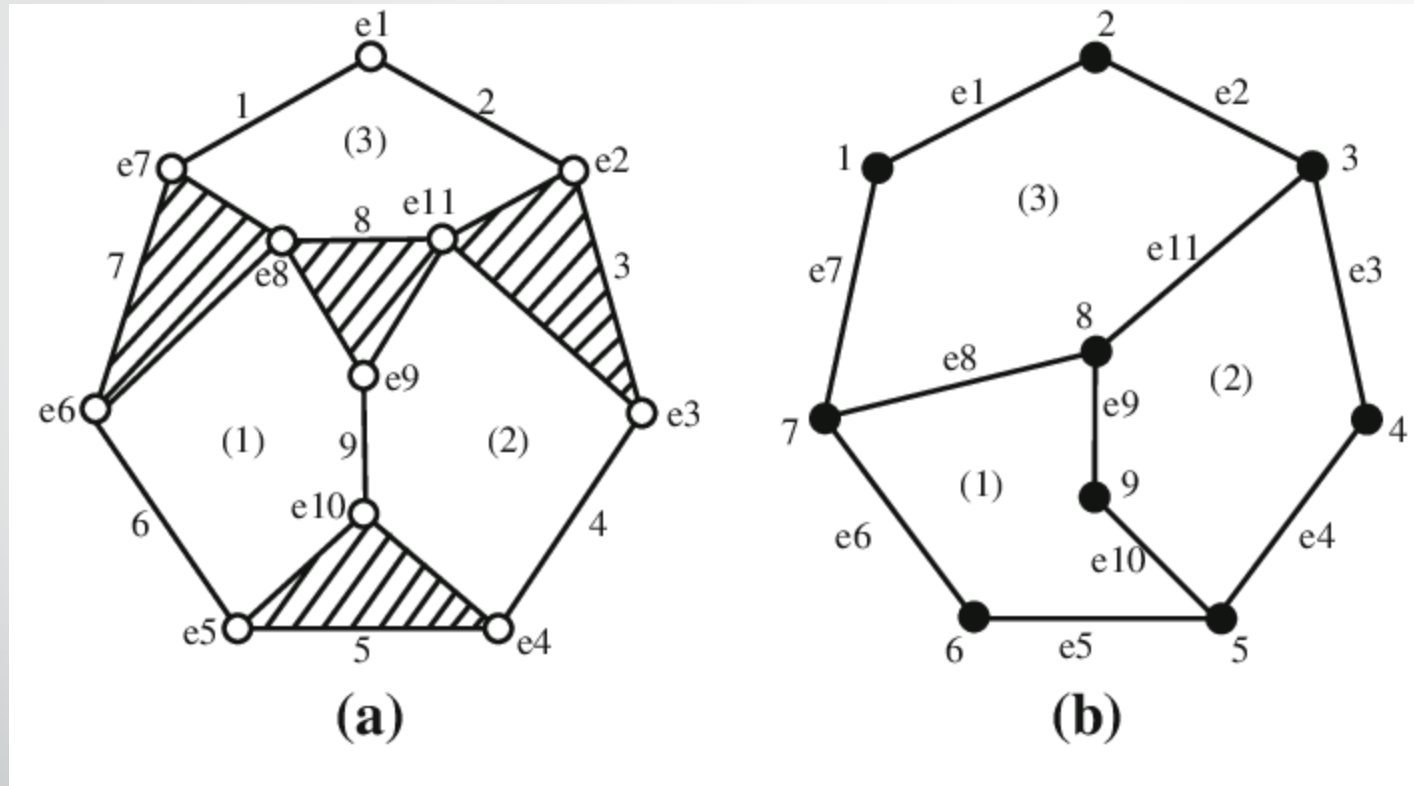
Therefore, it is an un-constrained chain.

Example 4: A eight-bar chain is shown here. Prove that it is an unconstrained chain.



Reference: researchgate.net

Example 5: A nine-bar chain is shown here. Prove that it is an unconstrained chain.



Reference: researchgate.net

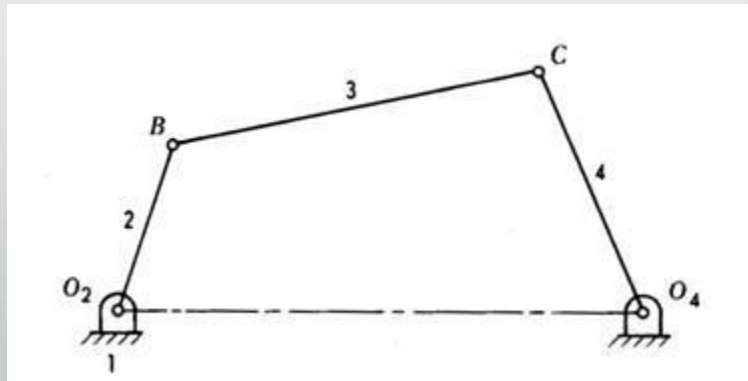
Mechanism:

When one of the links of a kinematic chain is fixed, the chain is called a mechanism.

Types of mechanisms:

Simple mechanism: a mechanism which has four links.

- In a simple machine, there is one point of application for the effort and one point for the load to be lifted.
- Simple machines are: lever, screw jack, inclined plane, bicycle.

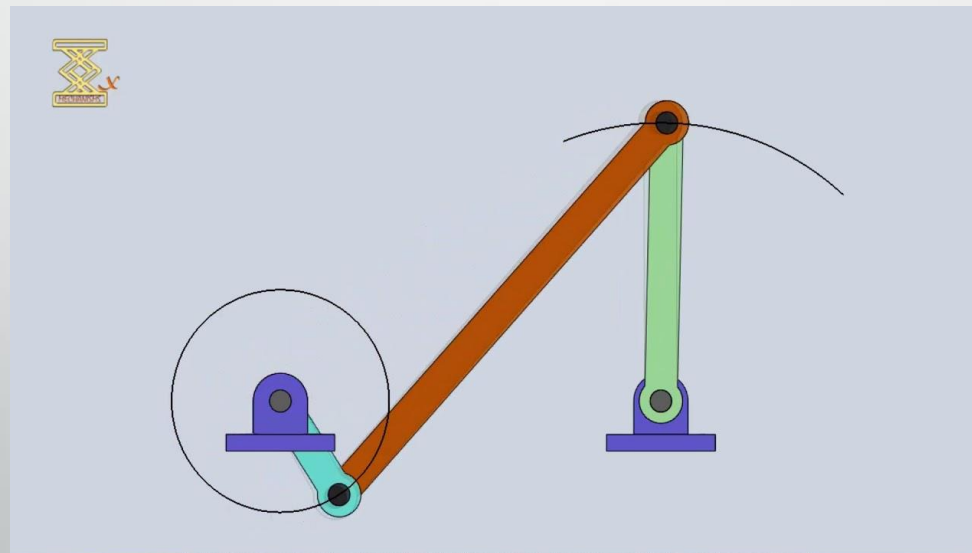


Reference: engineering.myindialist.com

Compound mechanism: a mechanism which has more than four links.

- In a compound machine, there are more than one point of application for the effort and the load.
- It may be thought of as a combination of many simple machines.

Compound machines are: Lathe machine, Grinding machine, Milling machine, Printing machine.



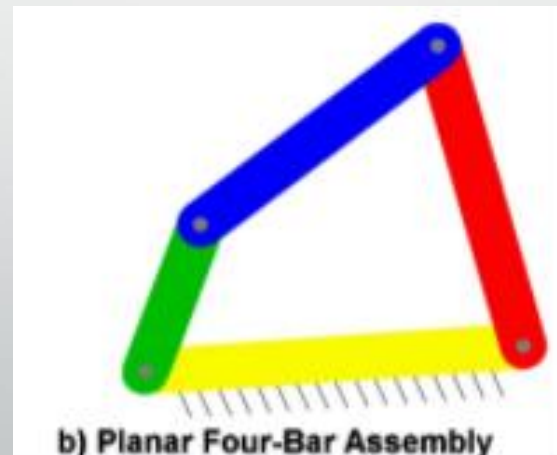
Reference: youtube.com

Complex mechanism: it is formed by the inclusion of ternary or higher order floating link to a simple mechanism.



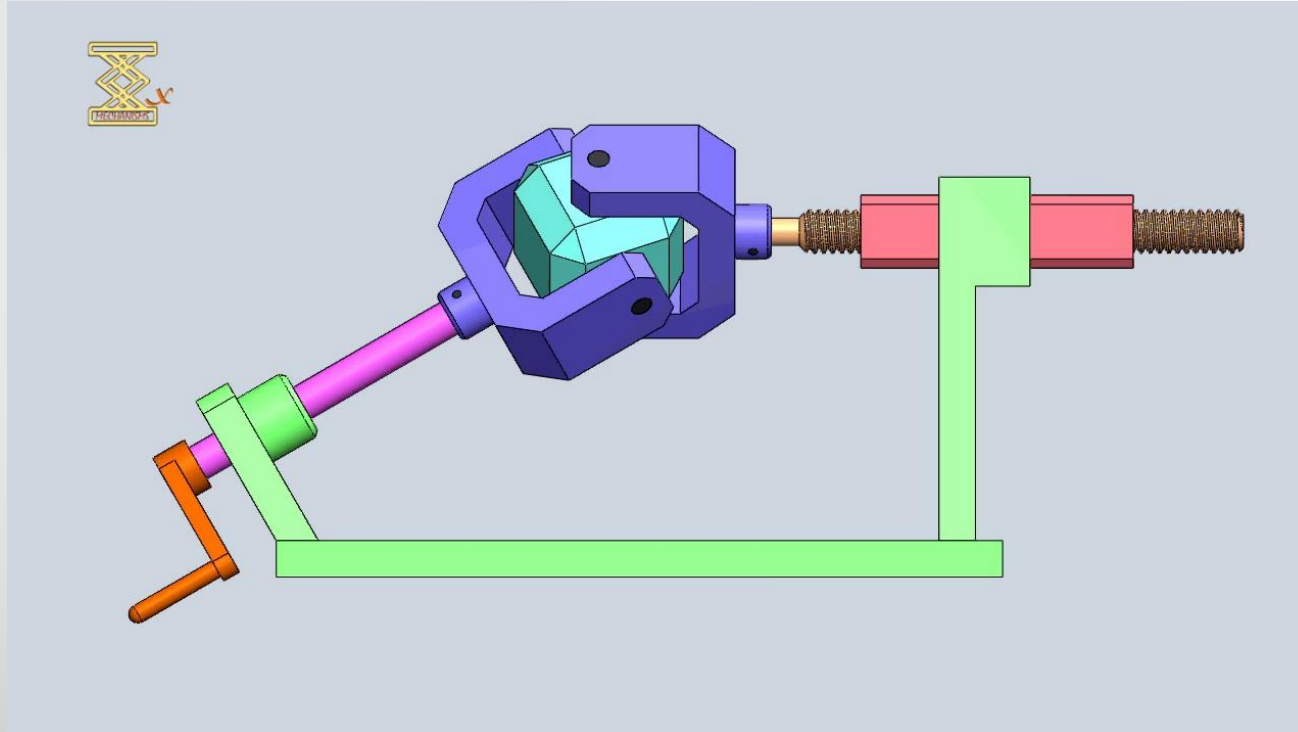
Reference: colourbox.com

Planar mechanism: when all the links of the mechanism lie in the same plane.



Reference: researchgate.net

Spatial mechanism: when the links of the mechanism lie in different planes.



Reference: youtube.com

Mechanism and Machines:

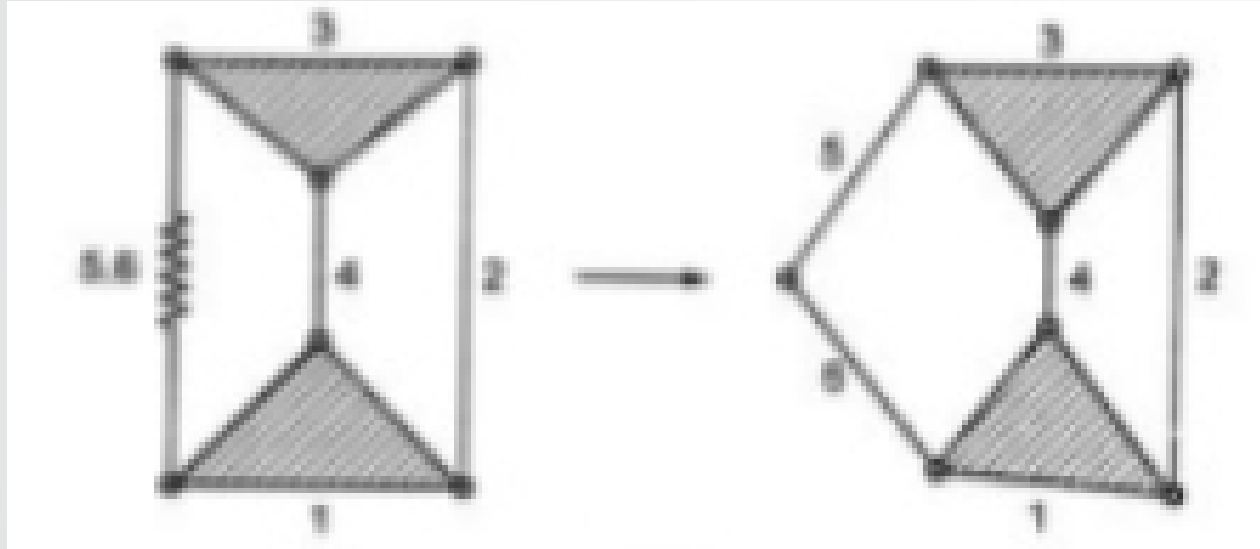
- A machine is a device that transforms energy available in one from to another to do certain type of desired useful work.
- Links may transmit both power and motion.
- A mechanism is a combination of rigid or restraining bodies, which are so shaped and connected that they move upon each other with definite relative motion.
- A mechanism is obtained when one of the links of the kinematic chain is fixed.
- A machine is a combination of two or more mechanisms arranged in such a way so as to obtain the required motion and transfer the energy to some desired point by the application of energy at some other convenient point.
- A machine is not able to move itself and must get the motive power from some source.

Equivalent Mechanisms:

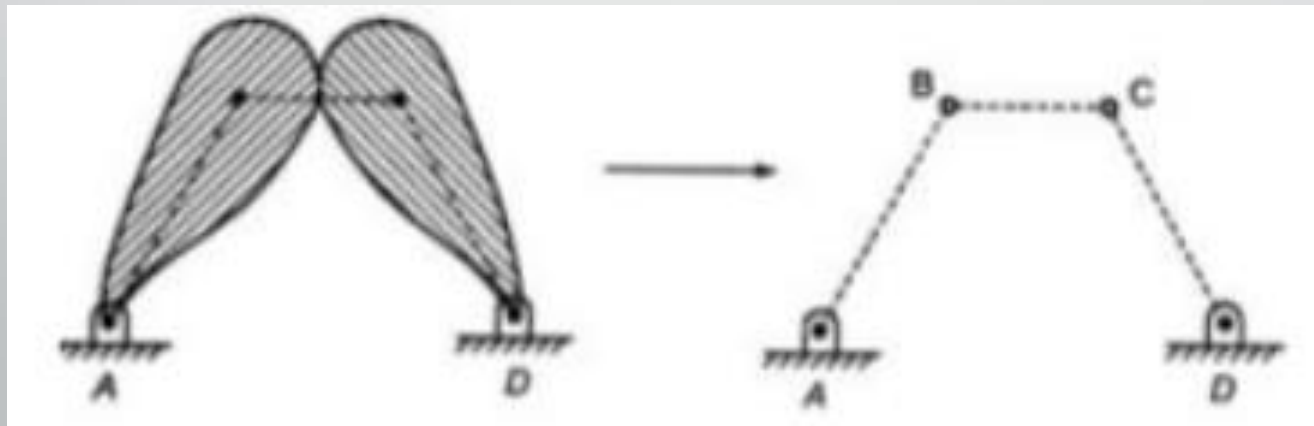
- Turning pairs of plane mechanisms may be replaced by other types of pairs such as sliding pairs or cam pairs.
 - New mechanism will be obtained having the same number of degree of freedom as the original mechanism is called the equivalent mechanism.
 - The equivalent mechanism will have same degree of freedom and shall be kinematically similar.
1. A sliding pair is equivalent to a turning pair.



2. A spring can be replaced by two binary links.



3. A cam pair can be replaced by one binary link together with two turning pairs at each end.



➤ Degree of freedom (DOF):

- ❖ An constrained rigid body moving in space can have three translations (u, v, w) and three rotational ($\theta_x, \theta_y, \theta_z$) motions about the three mutually perpendicular axes.
- ❖ Degree of freedom of a kinematic pair is defined as the number of independent relative motions, both translational and rotational.
- ❖ **Minimum number of independent coordinates (variables)** that need to be specified to fix the configuration of a mechanism.
- ❖ One link of the chain is grounded.

Degree of freedom = 6 – Number of restraints

The degree of freedom of some of systems are:

- Rigid body has six degree of freedom
- Rectangular bar sliding in a rectangular hole has one degree of freedom
- The Piston of the crank of a slider crank mechanism can be expressed by angular motion (one degree of freedom)
- A circular shaft rotating in a hole and also transmitting parallel to its axis has two degree of freedom
- A Ball and a socket joint has three degree of freedom

- Number of links = n_L
- Number of joints = n_J
- Degree of freedom of i^{th} joint = f_i

Degree of freedom of **planar mechanisms**:

$$F = 3(n_L - 1) - 3n_J + \sum_{i=1}^{n_J} f_i$$

Degree of freedom of **Spatial mechanisms**:

$$F = 6(n_L - 1) - 6n_J + \sum_{i=1}^{n_J} f_i$$

Degree of freedom (DOF)

- ❖ A mechanism must have at least 1 degree of freedom.
- ❖ A structure has zero degree of freedom.
- ❖ An over-constrained structure has negative degree of freedom.

Kutzbach criterion: The Kutbach criterion for determining the number of degree of freedom of a planar mechanism is:

$$F = 3(n - 1) - 2p - h$$

Here, F = degree of freedom

n = total number of links in a mechanism out of which one is fixed link

$n - 1$ = number of movable links

p = number of simple joints or lower pairs having one degree of freedom.

h = number of higher pairs having two degree of freedom or three degree of freedom



Thank You